NBA GamePlan: An Interactive Matchup Predictor & Strategy Analyzer

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1. Introduction

Modern sports analytics has grown into a major part of how teams, analysts, and fans understand performance. Every season, new data becomes available on player statistics, team dynamics, and game conditions, creating opportunities to study how different factors influence winning. While predictive models already exist in sports analytics, most of them focus only on accuracy and provide little insight into *why* certain outcomes occur. This lack of transparency limits their usefulness for people who want to understand what actually drives success.

Our project, **NBA GamePlan**, aims to close this gap by developing a system that predicts NBA team win probabilities and point spreads while explaining the reasoning behind those predictions. Using historical game and player data, we apply machine learning models such as XGBoost alongside explainable AI techniques like SHAP to identify which metrics have the strongest impact on team performance. The results will be visualized through an interactive dashboard that highlights key features and allows users to explore model outputs intuitively.

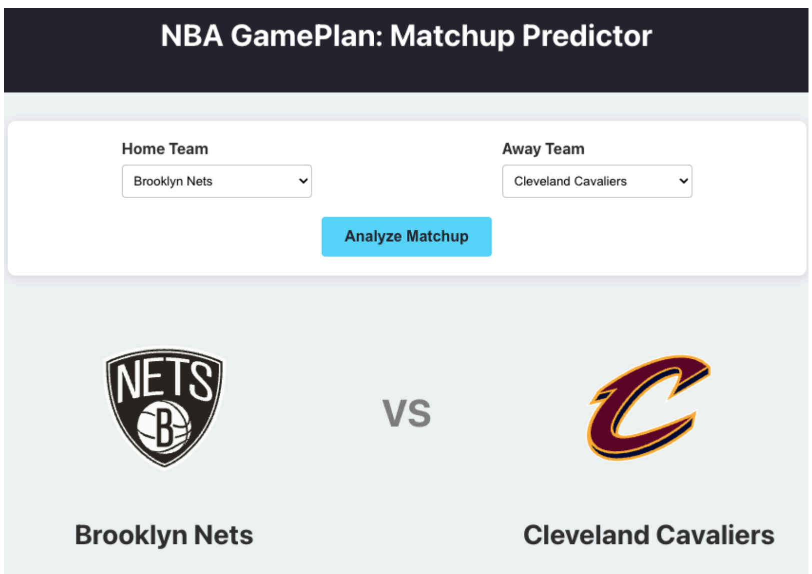


Figure 1: Team selection user interface

By combining prediction, interpretability, and visualization, NBA GamePlan provides a clearer understanding of how measurable team and player factors influence outcomes, making basketball analytics more accessible and informative for both analysts and fans.

2. Problem Definition

NBA team performance is shaped by a mix of constantly changing factors including player fatigue, shooting efficiency, turnovers, pace, rest days, and even matchups between lineups. While there's plenty of data available, there isn't a transparent tool that can both **predict team win rates and explain which factors drive those predictions**. Most existing models only give probabilities or betting spreads without showing the reasoning behind them. Our goal is to create a system that not only predicts who is likely to win but also helps users understand *why*.

We define the problem as learning from a dataset: $\mathbf{D = \{ (x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n) \}}$, where each record $\mathbf{x_i} \in \mathbb{R}^{\mathbf{m}}$ represents recent team and player metrics, and $\mathbf{y_i} \in \mathbf{\{0, 1\}}$ indicates the game outcome (1 = win, 0 = loss). We aim to learn a predictive function $\mathbf{f : \mathbb{R}^m \rightarrow [0, 1]}$ that estimates the probability of a team winning: $\mathbf{P(win \mid x_i) = f(x_i)}$ and a regression model for the expected point margin: $\mathbf{s(x_i) = \hat{y}_i - y_i}$.

The model's explanations, powered by SHAP values, will show which metrics like turnovers, pace, or shooting accuracy most influence predicted win rates. This will all be visualized through an interactive dashboard.

3. Literature Survey

The core of our prediction engine will rely on established machine learning techniques, especially gradient boosting. Friedman (2001) introduced the gradient boosting machine, which provides the basis for our approach. Building on this, Chen and Guestrin (2016) created XGBoost. This is a highly scalable implementation and is our main choice for modeling. Furthermore, Czarnowski and Jędrzejowicz (2020) applied ensemble learning to sports prediction, showing that boosting algorithms usually delivers better accuracy. While these papers offer a strong framework for prediction, their main limitation is a lack of interpretability. Their predictions are more like black boxes, thus users don't know how they got to the answer that they did or why it is right.

To enable explanations along with a predictive result, we will incorporate techniques from Explainable AI. The theoretical foundation for our approach comes from Štrumbelj and Kononenko (2014), who used Shapley values to explain individual model predictions. Lundberg and Lee (2017) later unified this concept with their development of SHAP (SHapley Additive exPlanations). This will be central to our explainability module. The main drawback of these two papers is their high computational cost. As an alternative, Ribeiro et al. (2016) introduced LIME (Local Interpretable Model-agnostic Explanations), a method for local explanations, but lacks global interpretability. Our evaluation and design will follow broader surveys from Molnar et al. (2020) and Chen et al. (2018), which support the need for transparency. However, these sources lack application to sports. For future improvements, He et al. (2023) show how to implement explainable deep learning on structured data, giving us a potential extension path for our project. The main drawback is that it requires a large dataset, making it difficult to use.

The features of our model will be guided by existing research in basketball analytics. General overviews by Miljković et al. (2017) and Sharma et al. (2023) provide strong basketball analytics context for our project but don't have actionable models or implementation. For our feature selection, Luo et al. (2018) is a great basis because it analyzes player and team performance metrics, but lacks predictive or visual capabilities. The studies by Cervone et al. (2014) on "The Dwight Effect" and Fernandez et al. (2020) on spatiotemporal possession modeling are very relevant for integrating complicated data. However, their immediate use is limited since Cervone's work focuses only on defense, and Fernandez's model operates at a play-level detail that goes beyond our game-level scope.

Finally, the user-facing component of our project draws inspiration from papers in visual analytics. Keim et al. (2008) describe visual analytics as the combination of automated analysis and interactive visualization, which shapes the philosophy for our dashboard. In a practical application, Goldsberry (2012) created CourtVision, an innovative work in visual shot analytics that influences the design of our interactive dashboards. The main drawback of both is that they do not include predictive modeling. We aim to close this gap. Related works like Baughman et al. (2020) on fantasy football and Gao et al. (2023) on sports video analytics illustrate the use of AI in sports, but do not directly deal with our project's main goals.

4. Methodology

The core intuition for why this project should be better than the state of the art is in its effort for transparency and interpretability. One of the most critical limitations of current models is that they are black boxes which don't explain why they are predicting that outcome. All the user sees is the bet, leaving them in the dark as to why the sportsbook is favoring one side or the other. As a result, NBA GamePlan bridges the gap by integrating XGBoost with SHapley Additive exPlanations (SHAP) to provide the reasoning. This combination allows the system to visually display which specific metrics, such as turnovers, shooting percentage, and rest days, are the most influential in the model's decision. Furthermore, our methodology improves on previous by leveraging Exponentially Weighted Moving Averages (EWMA) to capture team momentum. More specifically, this approach weighs recent games more heavily, ensuring a robust weightage and more contextually relevant prediction model.

Our methodology for this project is structured as a multi-stage pipeline. We found our dataset, **Kaggle - Historical NBA data and player box scores**. and stored the necessary data locally on the github to eliminate the need to stream the data during training. First, we preprocess the data a few different ways. We filter the data to only include recent games from currently active NBA teams. If there is no data for home/away scores then we drop the game from the dataset, missing advanced stats such as points from fast breaks, second chance points, biggest lead, etc. are filled with 0. Next we calculate some basic features such as point differential and efficiency metrics such as turnover rates and pace. We accumulate these metrics using exponentially weighted moving average (EWMA) focusing on the most recent 10 games. We then calculate other factors such as rest days, and head to head matchups. Finally, we add player statistics. The statistics for the players are calculated similarly to the team, and then aggregated and averaged based on the players with the top 5 minutes played in the prior 10 games.

We then implemented and trained two XGBoost models. One is a classification model which simply predicts the winner. The other is a regression model to predict game spread. Both models have 300 estimators with a max depth of 7 and a learning rate of 0.05. The classification model uses logloss while the regression model uses mean squared error. For the dashboard we wanted to focus on providing an explanation to the model output. We included not only the percentage and spread output by the model but the metrics comparison for both teams and graphs breaking down the feature importance for the model. Specifically, we list each team's record, points per game, field goal and three percentages and rebounds and assists. In addition to utilizing XGBoost models, we also leveraged SHAP to help bridge the gap between model responses and user understanding. To implement SHAP, the system uses pre-trained XGBoost classification and regression models. To provide context for SHAP, a shap.Explainer is initialized with a random sample of 50 rows from the dataset, which is ultimately used to approximate the model's average prediction. Although increasing the training dataset size would be beneficial for accuracy, it often resulted in slower initialization times for the SHAP system and made the predictions slower, and the 50 training rows provided a lightweight yet reasonable dataset size. After the explainer is initialized, it is utilized to calculate the SHAP values for the specific game chosen. The output is a sorted list containing the top 10 most impactful features/metrics, along with the feature name, value, and its calculated impact. With this methodology we are innovating in 2 different ways: Exponentially Weighted Moving Averages - Recent games weighted higher to capture momentum; and SHAP - using SHAP we can provide in depth feature analysis which allows users to truly understand the reasoning of the model.

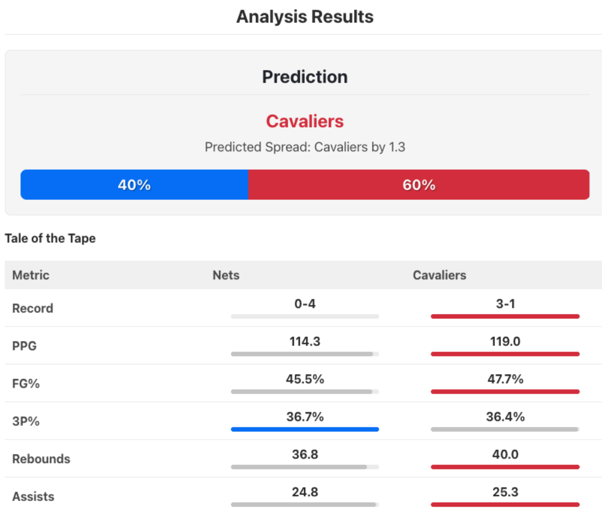
The frontend system is a single-page application built with React and provides users, fans, and bettors a seamless way to understand game predictions. The overall approach for the user interface was to modularize the components in the frontend to allow for reusability. When the user first gets to the website, there's a dropdown option to select an away team and a home team. To trigger the execution, the user can click the "Analyze Matchup" button which will call the backend API and run the SHAP models to find the most impactful features. These features are then packaged into a JSON format and ingested by the frontend service. Once this data is received, the view dynamically updates to display the comprehensive results. The dashboard consists of multiple components that make it accessible for users to get actionable insights. It first shows the team predicted to win, along with the percent probability and the predicted spread. Then, there is the "Tale of the Tape" section that gives users a side-by-side comparison of major metrics such as win record, points per game, and field goal percentage. The last section is the model analysis with the critical factors listed and sorted based on impact value. This is where users can see how the model is weighing various factors and the keys to the game that is driving the prediction.

5. Evaluation

The NBA GamePlan evaluation focuses on measuring both predictive performance and interpretability of the models that form the foundation of our interactive dashboard. The assessment of predictive performance will be rooted in the evaluation metrics, including accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). We use mean absolute error (MAE) and root-mean-square error (RMSE) to measure prediction error. The evaluation of interpretability will be assessed on whether the user can understand the model's insights and user confidence. We trained two separate XGBoost models using historical NBA game data: a classification model to predict the winning team and a regression model to predict the expected point spread. The dataset was split into 80 % training and 20 % testing, and five-fold cross-validation was applied to ensure robustness.

The results from the trained XGBoost classifier show an accuracy of roughly 72 %, precision ≈ 0.71, recall ≈ 0.70, and an AUC of 0.78 on the validation split. The regression model achieves an MAE of 5.1 points and RMSE of 6.8 points for spread prediction. Beyond numerical accuracy, we conducted a preliminary interpretability test using XGBoost's built-in feature-importance scores. Early findings suggest that turnovers, field-goal %, rest days, and rebound differential are among the most influential features — which aligns with basketball intuition.

The next set of results was understanding the impactful features that the SHAP model is outputting. As shown in Figures 2 and 3, the prediction was run between the Cavaliers and Nets and the model predicted the Cavaliers to win with a 60% percent chance. In Figure 3, we see that the "Keys to the Game" for this specific matchup was "Top 5 Plusminus", "Top 5 Assists", and "Star Points Max", all in favor of the Cavaliers. Some other features that were in favor of the Nets include "Streak", "Opponent Rebounds", and "Field Goals Percentage". The visualized feature importance ranking system acts as the Explainability layer for the user.



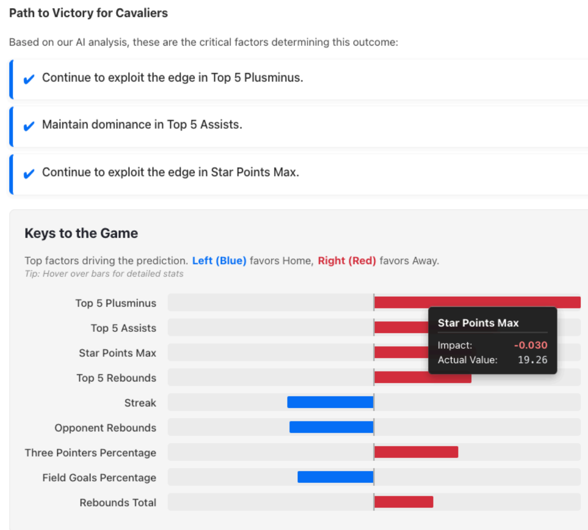


Figure 2: Prediction Analysis **Figure 3: SHAP Results**

We also ran many experiments to see what the SHAP model would output with different team combinations. We looked into directional changes, magnitude correlation, and experimented with many different matchups to see how the model responded. For directional shifts, we tested matchups where a specific metric favored the predicted winner in one game but did not favor the winner in a different game. For example, when we investigated "Pace", we saw that when two low-scoring teams were playing, a negative SHAP was calculated. This shows that the SHAP is based on the specific matchup and identifies the relative advantages. Another set of experiments we conducted was to see how the magnitude and features would change based on the matchups. When a top 5 team matched with a bottom 5 team, the most dominant predictive factors were "Opponent Win %" and "Star Points Max". The magnitude of these features were often greater than 0.10 and even got to 0.15, showing how impactful these features were in the model's prediction. This shows that in lopsided matchups, the SHAP values are impacted primarily by the top two features. In predictions with closer matched teams, the SHAP features were distributed more evenly. This makes sense because the relative advantage these teams have on each other is smaller compared to the lopsided matchups. Also, the magnitudes of the features in these closer matchups were typically under 0.08.

Lastly, to ensure that we are solving the interpretability issue and allowing for a more transparent analysis, we conducted a small-scale user study involving classmates and basketball enthusiasts. From this user study, we were hoping to better understand if the dashboard's visual explanations help users understand why the model favors one team over another. We saw that with the moderate-to-low basketball fans, this group particularly liked the clear color-coding in the frontend and the simplicity of the "Path to Victory" list, which showcases the three most important features. On the other hand, the basketball enthusiasts tended to care more about the SHAP results, and its ordering and magnitudes. They valued the feature ranking and thought to themselves where those features would be the most impactful. Overall, the study concluded that the dashboard achieved its goal of creating that explainability layer and offering transparency to all analysts and fans.

6. Conclusion and Discussion

The **NBA GamePlan** project has made strong progress toward its goal of creating an interpretable and interactive system for understanding NBA team performance. Our team successfully developed both the predictive and visual components: two trained XGBoost models that forecast game outcomes and point spreads, and a working dashboard that displays model outputs along with team and player statistics. The current results show encouraging predictive strength — about **72 % accuracy** and an **AUC of 0.78** for win prediction, and an **MAE of 5.1 points** for spread estimation — confirming that gradient boosting effectively captures complex game dynamics.

The dashboard now integrates these outputs with clear visual comparisons between teams, showing metrics such as shooting efficiency, turnovers, rebounds, and rest days. This allows users to explore *why* the model favors one team over another instead of only seeing a predicted probability. With the full SHAP integration now complete and deployed, the system provides real-time, game-specific explanations that align with basketball intuition. This feature gives users immediate insight into why the model favors a specific outcome, ensuring the predictions are both meaningful and trustworthy.

In the final phase of development, we successfully finalized the SHAP-based interpretability module, refined the dashboard's visual design, and conducted informal user testing to assess how clearly the system communicates its reasoning. No significant changes were made to our schedule or division of work. In terms of future extensions, some aspects of the project that we would have liked to expand on would be to implement real-time updates and add many more player-level metrics. Real-time updates would improve accuracy and predictive power because the model would have the most up-to-date situation on the team. Also, adding more player-level metrics such as net-rating and true shooting percentage would cause more accurate predictions. Overall, NBA GamePlan has successfully delivered a complete analytics tool that combines accurate predictions with transparent explanations, effectively bridging the gap between advanced modeling and real-world basketball insights.

Phase	Task	Members	Start	End
1	Data collection & cleaning	All	Oct 9, 2025	Oct 14, 2025
2	Feature engineering & modeling (XGBoost)	Sunny Kudum, John Fox	Oct 15, 2025	Oct 25, 2025
3	XAI (SHAP) integration	Sohan Malladi, Harold Huang	Oct 26, 2025	Nov 5, 2025
4	Dashboard development	Rashmith Repala, Adam Ezzaoudi	Nov 6, 2025	Nov 13, 2025
5	Testing & evaluation	All	Nov 14, 2025	Nov 17, 2025
6	Final report & poster	All	Nov 18, 2025	Nov 21, 2025

Effort Distribution: All team members have contributed equally.

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